LINUX ADMINISTRATION

FINAL PROJECT

PHASE I

Real-World Linux Systems Investigation, Professional Interview and Project Proposal

Assignment details	Requirement
Work type	Individual field investigation and written proposal
Main activity	Visit or contact a real organization and interview an IT professional
What to submit	One complete, professionally written PDF report
Where to submit	Google Form
Deadline	Sunday, 8 November 2026 at 5:00 AM (UTC+2)

DISCOVER | INTERVIEW | UNDERSTAND | PROPOSE

Read these instructions before contacting an organization or preparing your report.

Start Here: What You Need to Do

Your first task is to find out how Linux is used in a real workplace. Choose an organization, speak with an IT professional, learn about the Linux systems they use, and explain your findings in one well-organized report. Your report will help the lecturer choose the Linux project you will build in Phase II.

Step	Your action	Expected result
1. Find	Choose a real company or institution that uses Linux.	An appropriate organization and contact person.
2. Meet	Interview at least one person who works with its systems.	Real explanations from a knowledgeable professional.
3. Explore	Ask about the Linux tools, systems and projects in use.	An inventory of systems you are allowed to document.
4. Explain	Describe how the systems work and why they matter.	Clear technical and business explanations.
5. Propose	Suggest suitable projects you could build in a lab.	Up to three well-supported project choices.
6. Submit	Put everything into one PDF and upload it.	A complete report submitted before the deadline.

REMEMBER

Phase I is an investigation and proposal. You are not yet required to install, configure, or test the systems you discover. Wait for the lecturer to approve your Phase II project.

Key Dates and Rules

Deadline: Sunday, 8 November 2026 at exactly 5:00 AM, Rwanda time. This means early morning, not 5:00 PM.

Submission: Upload one PDF through the Google Form provided by the lecturer. Do not submit several separate files.

Interview: A real conversation with at least one relevant IT professional is compulsory. Internet research alone is not enough.

Safety: Do not access or test a company system unless you have separate written authorization. No such access is needed for this assignment.

1. Why We Are Doing This Project

Linux is used in many workplaces, often behind the services we rely on every day. It may run websites, databases, monitoring tools, cloud platforms, cybersecurity systems, or network services. Learning commands in class is important, but knowing why engineers use these tools is just as important.

This assignment gives you a chance to see how Linux supports real work. You will learn from professionals, study systems they use, and choose realistic ideas for your own practical project. You do not need to visit a large technology company. A smaller organization with a genuine Linux system is also suitable.

1.1 What You Should Learn

- Speak confidently and respectfully with working IT professionals.
- Recognize Linux-based solutions used in different IT roles.

- Explain a system in simple terms before discussing technical details.
- Connect a Linux solution to the problem it solves for the organization.
- Describe the main components, workflow, administration and security of a system.
- Identify a project you can safely build and demonstrate in a student lab.
- Present trustworthy findings in a clear academic report.

1.2 What This Assignment Does Not Ask You to Do

- Install or change software in the organization.
- Collect passwords, private data, secret configuration files, or customer records.
- Scan networks, test security weaknesses, or seek privileged access.
- Invent an interview or claim that a sample project is used by a company when it is not.

2. Choose a Real Organization

Select one organization where Linux supports at least one real business or technical activity. You may approach a bank, university, hospital, government office, software company, telecommunications company, ISP, nonprofit, retailer, manufacturer, or another organization with an IT team.

2.1 Who You May Interview

Ask for a professional who knows how the Linux system is used. Suitable people include a Linux administrator, systems engineer, network engineer, DevOps engineer, software engineer, cybersecurity analyst, cloud engineer, database administrator, IT operations specialist, or IT manager.

- Interview at least one relevant professional. Additional interviews are welcome if they help you understand different systems.
- A meeting at the workplace is preferred. A live online or telephone interview is acceptable when a visit is not possible.
- Explain that you are a student and that the discussion is for a university assignment.
- Ask permission before writing down a person's name, photographing equipment, or recording the conversation.

2.2 Record Basic Organization Details

- Organization name, or an agreed anonymous name if necessary.
- Type of organization and main services.
- City or district, if this information can be shared.
- Department or team responsible for the Linux system.
- Role of the person interviewed, interview date and meeting method.
- A short explanation of where Linux fits into the organization's work.

WHEN AN ORGANIZATION REQUIRES PRIVACY

You may describe it as, for example, "a financial institution in Kigali" and name the interviewee only by job title. Your report must still clearly explain the real systems and your interview process.

3. Prepare and Conduct the Interview

Prepare your questions before the meeting. Begin with broad questions and then explore individual systems. Let the engineer explain the work in their own words. Ask polite follow-up questions whenever something is unclear. Your goal is to understand the system, not to show off technical terms.

3.1 Recommended Interview Questions

About the engineer and the workplace

1. What is your role, and how do you work with Linux?
2. Which Linux distributions, tools or platforms does the organization use?
3. What important tasks depend on these systems?
4. Why did your team choose Linux or this particular solution?

About the systems

5. Which Linux systems or projects are currently in use?
6. What problem does each system solve?
7. Who uses it, and what does a normal day of use look like?
8. Could you explain the main parts of the system and how they work together?
9. How is Linux involved in running or managing the solution?
10. Which other services, applications or devices does it depend on?

About administration and support

11. How do engineers install, configure, update and monitor the system?
12. What routine tasks are automated?
13. What general security measures protect it?
14. How are backups and recovery handled, where relevant?
15. What common problems occur, and how does the team solve them?

About value and project ideas

16. What benefits has the system brought to your organization?
17. What would happen if it stopped working?
18. What would you improve if you had more time or resources?
19. Which part could a student safely reproduce using virtual machines or a test lab?
20. What skills and tools would you recommend for this type of project?

You do not need to use these questions word for word. Adjust them to fit the expert and the systems discussed. Ask for simple examples to help you explain each solution accurately.

3.2 Good Interview Practice

- Book the interview early and arrive on time.
- Introduce yourself and explain how the information will be used.
- Take clear notes. Confirm technical names and important points before you leave.
- Thank the professional for their time and respect any information they cannot share.
- Use your notes to write your own explanations. Do not submit a raw transcript as the main report.

4. What Linux Systems Should You Look For?

Linux is used in many areas of IT. The list below will help you ask useful questions and recognize possible projects. It is a guide, not a list of systems you must find. Your report must cover the real Linux systems that your chosen organization uses and allows you to discuss.

For every technology, confirm whether it actually runs on Linux or uses Linux as a major part of its environment. A tool being open source does not automatically make it Linux-based.

A. Linux servers and system administration

Examples: Ubuntu Server, Debian, Red Hat Enterprise Linux, Rocky Linux, AlmaLinux, SUSE Linux Enterprise, user and group management, SSH, Bash scripts, systemd services, updates and patching.

Ask about: How servers are installed, updated, secured, supported and used by the organization.

B. Network services and connectivity

Examples: BIND DNS, Unbound, Kea DHCP, FreeRADIUS, WireGuard, OpenVPN, Squid proxy, HAProxy, NGINX load balancing, nftables, Linux routing.

Ask about: How the services support communication, access, traffic control and network reliability.

C. Cybersecurity and security operations

Examples: Wazuh, Security Onion, Suricata, Snort, Zeek, Greenbone/OpenVAS, auditd, Fail2ban, Graylog, OpenSearch, Elastic Stack, SELinux, AppArmor.

Ask about: How engineers collect evidence, detect suspicious events, protect systems and respond to alerts.

D. Software development and web hosting

Examples: Apache HTTP Server, NGINX, PHP, Python, Node.js, Java, Apache Tomcat, GitLab, Gitea, web applications, internal APIs and development servers.

Ask about: How software is developed, tested, published and maintained on Linux.

E. DevOps, containers and cloud

Examples: Docker, Podman, Kubernetes with Linux nodes, Jenkins, GitLab CI/CD, Ansible, Terraform, Linux cloud virtual machines and deployment automation.

Ask about: How teams build, deploy, scale and automate services.

F. Databases and data services

Examples: PostgreSQL, MySQL, MariaDB, MongoDB, Redis, database replication, database backups and data processing platforms hosted on Linux.

Ask about: How information is stored, served, protected and recovered.

G. Monitoring, logs and operations

Examples: Zabbix, Nagios, LibreNMS, Prometheus, Grafana, syslog, rsyslog, Graylog, OpenSearch Dashboards and alerting tools.

Ask about: How teams know whether systems are healthy and how they investigate problems.

H. Storage, file sharing and backups

Examples: NFS, Samba, Ceph, MinIO, rsync, Restic, BorgBackup, Bacula and Linux-based backup or replication servers.

Ask about: How files and backups are shared, secured, restored and kept available.

I. Identity and access management

Examples: OpenLDAP, FreeIPA, Samba Active Directory, Kerberos, Keycloak, FreeRADIUS, centralized SSH login and access management.

Ask about: How users are identified, authenticated and given the right permissions.

J. Business, education and health applications

Examples: Moodle, Odoo, ERPNext, Nextcloud, OpenMRS, DHIS2, mail services such as Postfix and Dovecot, and Linux-hosted organizational portals.

Ask about: How Linux supports staff, students, customers, patients and everyday operations.

K. Virtualization and private infrastructure

Examples: KVM, QEMU, Proxmox VE, OpenStack, Linux hypervisors, virtual networking and automated virtual-machine deployment.

Ask about: How the organization runs many services efficiently and manages computing resources.

L. Special-purpose Linux systems

Examples: Raspberry Pi, embedded Linux, IoT gateways, research clusters, telecom services, industrial monitoring and other custom Linux solutions.

Ask about: What the system does, why Linux is useful and how engineers support it.

TECHNICAL ACCURACY

Some networking tools and firewalls are built on non-Linux operating systems. For example, pfSense and OPNsense use FreeBSD. Do not label them as Linux projects unless you are clearly describing a separate Linux component.

5. How to Explain Each System

A list of software names is not enough. The lecturer should understand what each system does, why the organization needs it, and whether it would make a good student project. Use the questions below as your writing guide.

Include	Explain in your own words
System name and role	What is the tool or service called? What job does it perform?
Problem and purpose	What need or problem led the organization to use it?
Linux environment	Which Linux distribution, server, container or virtual machine supports it, if known?
How it works	What are its main components? How does information or a request move through them?
Daily use	Who uses or manages it? What tasks happen regularly?
Administration	How is it configured, updated, monitored and supported?
Security and reliability	What general controls, logs, backups or recovery measures are used?
Benefits and limits	What is working well? What are the main challenges?
Student project idea	Could you build a small, safe version in a lab? What would you demonstrate?

5.1 Example: Move Beyond a Simple Statement

Too short: "The organization uses Zabbix to monitor servers."

Better: "The organization uses a Linux-hosted Zabbix server to track the health of selected servers and network services. Engineers review dashboards and alerts to find problems such as high resource usage or an unavailable service. The team uses monitoring to identify faults earlier and respond more quickly. A student could build a small test environment with one monitoring server and two monitored Linux machines."

The better description explains the purpose, normal operation, benefit and possible lab project. In your own report, use the details confirmed by the interviewee. Do not guess at the company's infrastructure.

5.2 Diagrams Are Welcome

A simple diagram can help explain how a system works. You may draw a high-level diagram showing a Linux server, application, users and data flow. Do not include sensitive addresses, confidential network layouts or internal security details without permission.

6. Build an Inventory of the Systems You Found

After the interview, organize all the Linux systems discussed into one inventory table. Give each system a number and use the same number when you describe it later in the report. If the organization uses many similar servers, you may group them by service or purpose while still explaining the important differences.

ID	System or project	Purpose	Linux or main tools
S01	System monitoring	Checks availability and performance	Ubuntu Server, Zabbix
S02	Internal web application	Serves an employee application	Linux, NGINX, PostgreSQL
S03	Log management	Collects system and security logs	Linux, Wazuh

These rows are examples only. Do not copy them as interview findings. Enter the real systems that the selected professional confirms.

How many systems? There is no fixed minimum beyond finding a genuine Linux use case. Investigate every relevant system the organization permits you to discuss. Quality matters more than a long list of names.

What if only one or two are available? Describe those systems well and explain why the organization could not share more. You can still propose a practical project based on a real use case.

7. Propose Projects for Phase II

Your report must not stop at describing existing systems. You must suggest how your findings could become practical Linux administration projects. Where possible, choose three candidate projects and rank them from your first choice to your third choice. If only one or two suitable projects are available, explain why.

7.1 For Each Candidate Project, Include

21. A clear project title that says what you will build.
22. The real problem or use case that inspired the idea.
23. The main objective and what the finished system should do.
24. The Linux distribution and software you plan to use.
25. The laboratory setup, such as virtual machines, containers or a small test network.
26. The main configuration tasks you expect to complete.
27. How you will test and demonstrate that the project works.
28. The resources, likely challenges and safety considerations.

7.2 Choose a Project That Is Practical

A strong project should teach useful Linux skills and be possible to complete within the time and resources provided by the lecturer. A smaller project that works well and can be explained clearly is better than an ambitious proposal with no realistic implementation plan.

Selection question	What to consider
Relevant?	Does it solve a real problem identified through your interview?
Useful skills?	Will you configure and administer meaningful Linux services?
Feasible?	Can you build it with available software, machines and time?
Safe?	Can it run in an isolated lab with no private organization data?
Demonstrable?	Can you show clear tests and results to the lecturer?

WHO APPROVES THE FINAL PROJECT?

The lecturer will review your report and decide which project you must implement in Phase II. The lecturer may approve, adjust, combine or replace a proposed idea.

8. Required Structure of Your PDF Report

Submit one well-organized document. Use the headings below in the same order so that the lecturer can quickly review your investigation.

Cover page: University name, course, assignment title, student name, registration number, program or class, report date and organization name if disclosure is allowed.

1. Executive summary: A short overview of the organization, the interview, the Linux systems found and your strongest project idea.

2. Introduction: Why you did the investigation, what you wanted to learn and how it relates to Linux Administration.

3. Organization profile: What the organization does, the team you spoke with and where Linux is used.

4. Interview and method: When and how the interview took place, the interviewee's role, the main discussion topics and how you checked the facts.

5. Linux systems inventory: One table listing every relevant Linux system you identified.

6. Detailed system descriptions: A separate explanation for each system, including purpose, Linux tools, workflow, administration, value and constraints.

7. Proposed projects: Up to three ranked project ideas with clear objectives, tools, lab plans and expected results.

8. Comparison and recommendation: Explain which candidate you prefer and why it is suitable for a student project.

9. Conclusion: Your main lessons from the interview and what the investigation taught you about Linux in practice.

10. References and appendices: Credit published sources and add permitted interview records, diagrams or supporting material.

9. Evidence, Permission and Academic Integrity

Your report must be based on honest fieldwork. It must also respect the people and organizations who help you. Good professional judgment is part of this assignment.

- Record the date, method and professional role for every interview.

- Do not include secrets, customer information, passwords, internal IP plans, sensitive diagrams or private source code.
- Do not ask for access to live servers. A demonstration provided by the engineer is optional and must follow the organization's rules.
- Do not invent interview answers or present sample technologies as if you observed them at the company.
- Do not paste an AI-generated report and claim it describes a real environment. You may use writing tools to improve grammar, but the findings and explanations must remain accurate and your own.
- Use public documentation to check technical names or concepts. Clearly separate that research from what you learned in the interview.

If a question cannot be answered because the information is confidential, write "Not disclosed" or explain the restriction briefly. Do not guess.

10. Submission Instructions

When your report is complete, export it as a single PDF file. Open the PDF and check that all pages, tables, diagrams and references are visible. Then submit that one file using the Google Form provided by the lecturer.

10.1 File Name

Use this naming format:

REGISTRATION_NUMBER_LINUX_FINAL_PROJECT_PHASE1.pdf

Replace REGISTRATION_NUMBER with your own student registration number. Do not use a sample number in your final file.

10.2 Deadline

SUBMISSION CLOSES

Sunday, 8 November 2026 at 5:00 AM (CAT, Rwanda time, UTC+2). This is 5:00 in the morning.

- Submit through the lecturer's Google Form, not by email, unless the lecturer gives different instructions.
- Upload one PDF containing the full report and any permitted appendices.
- Make sure you select the correct final PDF and confirm that Google Forms accepted the submission.
- Plan ahead. Problems with internet access or file conversion can take time to solve.

Late submissions will be handled according to the lecturer's and institution's assessment rules.

11. What Happens After Submission?

Once all reports are submitted, the lecturer will review the systems and project ideas in each report. The lecturer may choose one of your proposed projects or give you a revised project that meets the course requirements. You will then receive the instructions for Phase II.

Phase II Will Focus on Practical Work

- Preparing a safe Linux test environment.
- Installing and configuring the approved Linux services.
- Managing users, access, security, networking, monitoring or other relevant functions.
- Testing your configuration and fixing problems.
- Presenting evidence that the system works and explaining your decisions.

Appendix A. Interview Record Template

Field	What to record
Organization	Name or permitted anonymous description
Industry	Main field of work
Interview date	Day, month and year
Interview method	In person, video call or telephone
Professional interviewed	Name with permission, or job title only
Professional role	Main areas of technical responsibility
Main Linux technologies	Short list confirmed during the discussion
Important lessons	Three to five key ideas from the interview
Permission or restrictions	What may be named, shown or quoted in the report

After this table, add a short summary of the interview in your own words. You may attach selected questions and answers when useful, but a verbatim transcript is not required.

Appendix B. Linux System Profile Template

Create one profile for every significant Linux system identified. You may use shorter profiles for minor systems.

Field	Details to complete
System ID and name	For example, S04, [System name]
Purpose	The problem it solves and who uses it
Linux environment	Distribution, virtual machine, container or host, if known
Main components	Relevant software, services and integrations
How it works	A plain-language explanation of its normal workflow
Administration	Updates, configuration, logs, monitoring and routine tasks
Benefits and challenges	What helps the organization and what remains difficult
Lab project possibility	What a student could safely build and test
Source of information	Interview notes, approved demo or public documentation

Appendix C. Project Proposal Template

Use the following short format for each of your preferred projects.

Proposal item	What to write
Project rank and title	Choice 1, Choice 2 or Choice 3; use a specific title
Problem to solve	Explain the need observed or discussed in the interview
Project objective	Describe the result you want to achieve
Linux tools	List the expected distribution and main software
Lab design	Describe the virtual machines, containers or test devices needed
Implementation activities	Identify the main tasks you would configure
Testing plan	How you would prove that the solution works
Resources and risks	Time, computing resources and challenges
Reason for choosing it	Explain its value for learning Linux Administration

Appendix E. Simple Project Comparison

Use this table to support your final project recommendation. Rate each factor as High, Medium or Low, and briefly explain your ratings.

Factor	Candidate 1	Candidate 2	Candidate 3
Relevance to real need			
Linux skills gained			
Ease of building in a lab			
Safety and independence from private data			
Ease of testing and demonstration			
Overall recommendation			

Appendix F. Final Student Checklist

Review each item before uploading your PDF.

- ☐ I identified a real organization where Linux is used.
- ☐ I interviewed at least one relevant IT professional.
- ☐ I recorded the interview date, method and professional role.
- ☐ I listed all relevant Linux systems the organization allowed me to discuss.
- ☐ I explained what each system does, how it works and why it is used.
- ☐ I described the benefits, challenges and administration involved.
- ☐ I included up to three realistic project ideas, or explained why fewer are available.
- ☐ I clearly recommended the project I believe is most suitable.
- ☐ I checked technical accuracy and cited public sources where needed.
- ☐ I removed confidential material and used names or photos only with permission.
- ☐ I named the file correctly and submitted one PDF through **Google Forms** before **Sunday, 8 November 2026 at 5:00 AM (CAT)**

FINAL MESSAGE

Choose a real environment. Ask good questions. Explain what you learned in your own words. The strongest project proposal is one that is clear, useful, technically correct and possible to build safely.